Design Document Project 3: FactOrNot

Project Description:

FactOrNot is a community-driven fact-checking platform where users submit claims, news headlines, or viral statements, and the community collectively verifies them as "Fact" or "Not." Each submission gets a credibility score based on community votes, and users can add evidence-based comments (with source links) to support or debunk a claim. Posts are organized under topic channels (Politics, Health, Tech, Science, etc.) so users can browse and verify claims by category.

Unlike traditional social media where misinformation spreads unchecked, FactOrNot puts the verification process front and center. Every claim displays a real-time credibility meter showing the community's consensus, the number of verifications, and the ratio of Fact vs. Not votes. Claims with the most engagement and the most contested credibility scores trend to the top — surfacing the statements that need the most scrutiny.

All verification is anonymous to reduce bias and encourage honest assessment. The platform promotes media literacy and critical thinking by making fact-checking interactive, accessible, and community-powered.

User Personas:

Persona 1 — Derek, The Social Media Scroller

Age: 21 **Occupation:** Junior, Computer Science, Boston University **Location:** Boston, MA **Tech Comfort:** High

Background

Derek spends about four hours a day on Instagram, Reddit, and X. He constantly sees viral headlines and shocking statistics shared by friends and influencers. He usually reshares things that feel true without checking them, but recently got called out publicly for sharing a false

health claim. That embarrassed him and now he wants a quick, reliable way to verify things before he hits repost.

Goals

- Quickly check whether a viral claim is true before sharing it
- See what other people think about a claim without reading long articles
- Contribute his own verdict to help the community

Frustrations

- Fact-checking websites like Snopes take too long to read
- He cannot always tell if a source is credible
- Most fact-checkers feel politically biased to him

How he uses FactOrNot

Derek opens the app on his phone when he sees something suspicious on social media. He searches for the claim in the Health or Politics channel, looks at the credibility score, reads a few evidence comments, and makes up his mind in under two minutes. If the claim is not already there he submits it himself so the community can weigh in.

Persona 2 — Mia, The Investigative Journalist

Age: 34 **Occupation:** Digital Reporter, regional news outlet **Location:** New York, NY **Tech Comfort:** High

Background

Mia covers misinformation and media literacy for a mid-size digital news publication. She regularly monitors how viral claims spread online and wants to understand public perception before writing a story. She is frustrated that most platforms amplify false claims instead of flagging them. She sees FactOrNot as a tool to gauge community consensus and contribute professionally sourced rebuttals that are more accessible than a full article.

Goals

- Monitor which claims are gaining traction in the community
- Add high-quality evidence comments with links to primary sources

- Use community credibility scores as data points for her reporting

Frustrations

- Social media platforms do not show her aggregate public opinion on specific claims
- Her corrections never reach the same audience as the original false post
- Evidence-based rebuttals get buried under low-effort replies on other platforms

How she uses FactOrNot

Mia checks the Trending sort in the Media and Politics channels every morning. When she spots a claim she has researched, she adds a detailed evidence comment with a link to a peer-reviewed study or official source. She also submits new claims she encounters in her reporting so the community can react before her article is published.

Persona 3 — Raj, The Critical Thinker

Age: 28 **Occupation:** Data Analyst, financial firm **Location:** Chicago, IL **Tech Comfort:** Very High

Background

Raj studied philosophy and statistics and is deeply interested in epistemology — how we know what we know. He distrusts media on all sides and prefers to evaluate claims based on evidence and logic rather than authority. He loves debate and regularly engages on Reddit and in online forums, but finds those platforms too chaotic and unstructured. He wants a focused environment where the single question on the table is always "is this true or not."

Goals

- Browse claims by topic and evaluate them methodically
- Read evidence from both sides before casting a verdict
- See how his assessment compares to the community consensus
- Identify which claims are most contested and therefore most interesting

Frustrations

- Online debates get derailed by personal attacks and off-topic replies
- There is no clean way to see a summary of the strongest arguments on each side
- Most platforms reward outrage over accuracy

How he uses FactOrNot

Raj spends time in the Science and Economy channels sorting by Contested to find claims where the community is most divided. He reads every evidence comment carefully before voting. He contributes detailed neutral or fact-supporting comments with links to data sources. He checks back on claims he voted on to see if the consensus shifted after more votes came in.

Persona 4 — Zoe, The Concerned Parent

Age: 41 **Occupation:** Elementary School Teacher **Location:** Austin, TX **Tech Comfort:** Medium

Background

Zoe is a mother of two children aged 7 and 10. She is active in parent Facebook groups and school newsletters where health and safety claims circulate constantly — vaccine side effects, food additives, screen time studies, and school policy claims. She is not a researcher and does not have time to dig through medical journals, but she takes her children's health seriously and wants a simple, trustworthy way to verify things she reads before acting on them or sharing them with other parents.

Goals

- Quickly verify health and safety claims before making decisions for her family
- Find a platform that is easy to use without a technical background
- Trust that the community verdicts come from real reasoning not just opinions

Frustrations

- She cannot tell the difference between credible and non-credible websites
- Parent groups on Facebook spread misinformation constantly with no correction mechanism
- Most fact-checking tools use jargon that is hard to understand

How she uses FactOrNot

Zoe goes straight to the Health and Health Myths channels when she encounters a claim in a parent group. She looks at the credibility meter and reads the top evidence comments, paying special attention to comments with source links. She votes based on what the evidence

comments tell her. Occasionally she submits claims she cannot find already in the system. She does not write long comments but appreciates when others do.

User stories (use cases but with a story)

Kunal Takke: Claims & Verification

• As a user, I want to submit a claim or headline for the community to verify so we can collectively determine its credibility.
• As a user, I want to browse all claims sorted by credibility score so I see the most contested and actively verified claims first.
• As a user, I want to vote "Fact" or "Not" on a claim to contribute my assessment to the community verdict.
• As a user, I want to add an evidence comment with an optional source link to support my verdict with reasoning.
• As a user, I want to filter claims by topic/channel so I can focus on areas I care about (e.g., Health, Politics, Tech).
• As a user, I want to edit my submitted claim if I need to correct or clarify the wording.
• As a user, I want to delete my submitted claim if it is no longer relevant.

Kunal Juvvala: Topics & Channels

• As a user, I want to view a list of available topic channels organized by category.
• As a user, I want to create a new channel for a topic area that doesn't exist yet.
• As a user, I want to click on a channel and see all the claims inside it.
• As a user, I want to edit a channel's name or description.
• As a user, I want to delete a channel that is no longer relevant.

Design Mockups:

The following wireframes illustrate the layout and structure of each page in FactOrNot. The application consists of the following pages.

Mockup 4: Login & Register Pages

[✓] FactOrNot Claims Channels

Login | Register

Login Screen

Login to FactOrNot

Verify claims. Fight misinformation.

Username

Enter your username

Password

Enter your password

Login

Don't have an account? Register here

Demo account: **demo** / **demo**

Register Screen

Create Account

Join the community. Fight misinformation.

Username

Choose a username

Password

At least 4 characters

Confirm Password

Confirm your password

Register

Already have an account? Login here

Mockup 1: Claims Feed (Home Page)

[✓] FactOrNot Claims Channels

Login | Register

Claims Feed

+ Submit Claim

Search claims... | Search

All Channels ▼ | Sort by Credibility ▼

1050 claims found

Health — 01/15/2025

Drinking 8 glasses of water daily is essential for health

Likely Fact — 72%

144 Fact — 200 votes — 56 Not

by truthseeker42 — 3 evidence

Technology — 02/03/2025

AI will replace 40% of jobs within the next decade

Contested — 48%

89 Fact — 185 votes — 96 Not

by skepticalmind — 5 evidence

Science — 12/20/2024

Brain cells continue to grow throughout entire human lifespan

Likely False — 31%

40 Fact — 129 votes — 89 Not

by factchecker_pro — 2 evidence

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Mockup 2: Claim Detail Page

[✓] FactOrNot Claims Channels

Hi, demo | Logout

← Back to Claims

Health

Submitted 01/15/2025 by **truthseeker42**

Drinking 8 glasses of water daily is essential for health

This widely circulated health recommendation has been repeated for decades, but is there actual scientific evidence to support the specific 8-glass threshold?

View Original Source

Likely Fact 72%

144 Fact | 200 votes | 56 Not

✓ Fact | ✗ Not

Edit Claim | Delete Claim

Evidence & Comments (3)

+ Add Evidence

Your Analysis

Share your evidence or reasoning...

Source URL (optional)

https://...

Verdict

Supports the claim (Fact) ▼

Submit Evidence

SUPPORTS **factchecker_pro** 01/16/2025

I found a peer-reviewed study that supports this claim. Hydration levels correlate with improved cognitive function and physical performance.

View Source

DEBUNKS **skepticalmind** 01/17/2025

The original "8 glasses" recommendation has no scientific basis. A 2002 review in the American Journal of Physiology found no evidence supporting this specific amount.

View Source

SUPPORTS **datadriven** 01/18/2025

While the exact number varies, adequate hydration is confirmed essential. The National Academies recommends 3.7L daily for men and 2.7L for women from all sources.

Mockup 3: Topic Channels Page

[✓] FactOrNot Claims Channels Hi, demo Logout

Topic Channels

+ Create Channel

Browse claims by topic. Click a channel to see its claims.

P **Politics**
POLITICS

Political claims, government policies, and election news

105 claims by truthseeker42

H **Health**
HEALTH

Medical claims, wellness trends, and health misinformation

105 claims by factchecker_pro

T **Technology**
TECHNOLOGY

Tech industry claims, product rumors, and digital trends

105 claims by datadriven

S **Science**
SCIENCE

Scientific discoveries, research claims, and environmental topics

105 claims by logicmaster

F **Finance**
FINANCE

Economic claims, market predictions, and financial advice

105 claims by criticalthink

Edit Delete

E **Entertainment**
ENTERTAINMENT

Celebrity news, movie rumors, and pop culture claims

105 claims by newswatch

Screen 7 – Login

```
FactOrNot                                   [ Channels ]  [ Sign In ]

                  FactOrNot
                  Sign in to submit claims and add evidence

                  ! "No account found with that username."
                  (error shown on failure)

                  Username
                  your_username

                  Password
                  ........

                  Sign In

                  No account? <Create one>

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```

Notes

- Passport.js local strategy validates username and bcrypt hash
- On success: 7-day session cookie stored in MongoDB, user redirected home
- On failure: error message shown inline
- "Create one" switches to the Register view (Screen 8)